PHYB54: Lecture 10
Oscillations

Chapter-5

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5.1 Hooke's Law

- As you know, Hooke's Law (spring law) gives the force due to a spring, in the form (assuming it is along the x axis):
$$F_x(x) = -kx$$
where x is the displacement relative to the spring's equilibrium point. The force can be in either direction—in the $+x$ direction when x is negative, and in the $-x$ direction when x is positive (at least for the case when k is positive).
- Since the force is conservative, we can write the force as minus the gradient of a potential energy $U(x) = \frac{1}{2}kx^2$.
- From what we have **learned about potential energy graphs, you can immediately see that this upward curving parabola indicates** that $x = 0$ **is a stable equilibrium** (if k were negative, the parabola would be curved downward and $x = 0$ would be an unstable equilibrium).
- However, **Hooke's Law has a general validity for the following reason**: for any general potential energy curve $U(x)$ in the vicinity of an equilibrium point at $x = x_o$, which we can take to be zero, we can always perform a Taylor-series expansion about that point:
$$\boxed{U(x) = U(0) + U'(0)x + \tfrac{1}{2}U''(0)x^2 + \cdots.}$$

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Hooke's Law-2

- The first term is the constant term, but since potential energy can be defined with any zero point, the constant term can be ignored (or considered to be zero).
- The second term is the linear term, but since $x = 0$ is an equilibrium point, the slope there is by definition zero, so the first non-zero term is the third term:

$$U(x) = \tfrac{1}{2}kx^2.$$

- This says that for sufficiently small displacements from an equilibrium point, Hooke's Law is ALWAYS valid for any potential energy function. This justifies our consideration of this case in detail.
- Of course, Hooke's Law can relate to any coordinate, not just x. Let's revisit the box on a cylinder problem as an illustration.

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Example 5.1: Cube Balanced on a Cylinder

- Statement of the problem:
  - Consider again the **cube of Example 4.7** and show that for small angles θ the potential energy takes the Hooke's Law form $\boldsymbol{U(\theta) = \tfrac{1}{2} k\theta^2}$**.**
- Solution:

$$U(x) = U(0) + U'(0)x + \tfrac{1}{2}U''(0)x^2 + \cdots.$$

  - We found before that the potential energy for the cube in terms of θ was

$$\boxed{U(\theta) = mg[(r+b)\cos\theta + r\theta\sin\theta].}$$

  - For small θ, we can make the approximations $\cos\theta \sim 1 - \theta^2/2$; $\sin\theta \sim \theta$. Notice that we kept the TWO leading terms for $\cos\theta$. Why? With these substitutions, we have

$$U(\theta) \approx mg[(r+b)(1-\tfrac{1}{2}\theta^2) + r\theta^2] = mg(r+b) + \tfrac{1}{2}mg(r-b)\theta^2.$$

  - Ignoring the constant term, we see that this is in the form of Hooke's Law, with a "**spring constant**" $\boxed{k = mg(r-b).}$
  - Notice that the **equilibrium is stable** only when $k > 0$, i.e. when $\boxed{r > b}$ as we found after quite a bit more work before.
- Notice also that since $U(x) = \tfrac{1}{2}kx^2$ is a parabola, the turning points (at least for small displacements) are equidistant from the equilibrium point.

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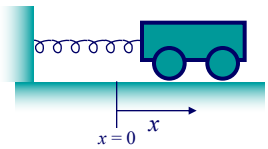
5.2 Simple Harmonic Motion

- Let's now look at all of this from the point of view of the equation of motion. Consider a cart on a frictionless track attached to a spring with spring constant k. Since we have

$$F_x(x) = -kx$$

the equation of motion is

$$\boxed{m\ddot{x} = -kx} \quad \Rightarrow \quad \ddot{x} = -\frac{k}{m}x = -\omega^2 x$$



where we introduce the constant

$$\omega = \sqrt{\frac{k}{m}}$$

which represents the angular frequency with which the cart will oscillate, as we will see.

- Because Hooke's Law always applies near equilibrium for any potential energy, we will find oscillations to be very common, governed by the general equation of motion such as for a pendulum vs. angle ϕ:

$$\boxed{\ddot{\phi} = -\omega^2 \phi}$$

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Exponential Solutions

- The equation $\ddot{x} = -\omega^2 x$ is a second order, linear, homogeneous differential equation. Therefore, it has two independent solutions, which can be written

$$\boxed{x(t) = e^{i\omega t} \quad \text{and} \quad x(t) = e^{-i\omega t}.}$$

- As you can easily check, both of these solutions do satisfy the equation. However, you should have come to expect two arbitrary constants in the solution to a second-order differential equation (two constants of integration), and in fact any linear combination of these two solutions is also a solution $\boxed{x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}.}$

  This is called the superposition principle, which works for any linear system.
- Any solution containing two arbitrary constants is in fact the **general** solution to the equation.
- This solution can be written in terms of Sine and Cosine by using Euler's equation $e^{\pm i\omega t} = \cos(\omega t) \pm i\sin(\omega t)$.
- Plugging these into our original solution, we can write

$$x(t) = (C_1 + C_2)\cos(\omega t) + i(C_1 - C_2)\sin(\omega t) = \boxed{B_1 \cos(\omega t) + B_2 \sin(\omega t)}$$

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Sine and Cosine Solutions

- Thus, the solutions $x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$, and $x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$, are equivalent so long as

$$B_1 = C_1 + C_2, \quad \text{and} \quad B_2 = i(C_1 - C_2).$$

- An important **issue is that $x(t)$, being a actual position coordinate**, has **to be real**. In general, the first solution **looks as if it is complex**, while the **second solution looks as if it is real**. However, this depends on whether **the coefficients are real or not**. Both C_1 and C_2 can be complex, but if both their sum and difference $C_1 + C_2$ and $C_1 - C_2$ are real then B_1 and B_2 are real and in **fact both versions are completely equivalent**. This solution is the definition of **simple harmonic motion (SHM)**.
- As usual, we **determine these constants from the initial conditions.**
- **If I start the motion by pulling the cart aside (position $x(0) = x_o$) and releasing it ($v(0) = 0$) then only the cosine term survives** and

$$\boxed{x(t) = x_o \cos(\omega t).}$$

- **If the cart starts at $x(0) = 0$ by giving it a kick (velocity $v(0) = v_o$) then only the sine term survives** and $\boxed{x(t) = \frac{v_o}{\omega} \sin(\omega t).}$

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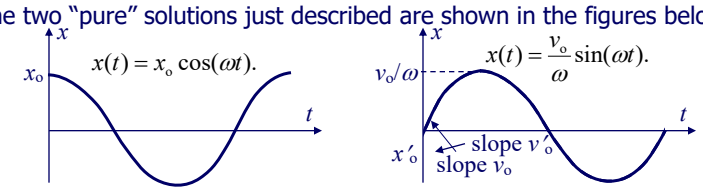
Phase Shifted Cosine Solution

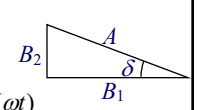
- The two "pure" solutions just described are shown in the figures below.



- The general solution, **if I both pull the cart to the side and give it a push (both x_o and v_o non-zero), the motion is harder to visualize**, but you may expect it to be the same type of motion, with some **phase shift.**
- We can demonstrate that for the general case by defining another constant

$$A = \sqrt{B_1^2 + B_2^2},$$

where A is the hypotenuse of a right triangle whose sides are B_1 and B_2. Obviously there is also an associated angle δ such that $B_1 = A \cos \delta$ and $B_2 = A \sin \delta$. Thus $x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$



$$= A[\cos\delta \cos(\omega t) + \sin\delta \sin(\omega t)]$$
$$= \boxed{A \cos(\omega t - \delta).}$$

Note that you have already solved this same problem using energy, in problem 4.28.

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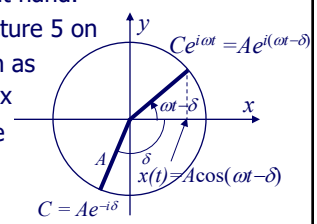
Solution as Real Part of $e^{i(\omega t-\delta)}$

- Notice that in $x(t) = A\cos(\omega t - \delta)$, there are still two constants, A **and** δ**, that are to be determined by initial conditions.**
- But notice that we can rewrite this using **Euler's equation** as $x(t) = A\cos(\omega t - \delta) = \mathrm{Re}\left[Ae^{i(\omega t-\delta)}\right]$, where Re[] denotes the real part of what is in the square brackets. As a practical matter, then, we can write the solution in terms of a complex exponential, and after any mathematical manipulations, simply throw away any complex part and keep the real part of the solution as the answer to the physical problem at hand.
- Graphically, if you will recall our discussion from Lecture 5 on complex exponentials, we can represent the solution as the x (real) component, or projection, of the complex number represented by a point in the complex plane given by $Ae^{i(\omega t-\delta)}$.
- You should train yourself to be able to go easily between the phase shifted cosine and Re[] complex representations.



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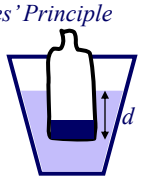
Example 5.2: A Bottle in a Bucket

- Statement of the problem:
  - A bottle is floating upright in a large bucket of water as shown in the figure. In equilibrium it is submerged to a depth d_o below the surface of the water. **Show that if it is pushed down to a depth** d **and released, it will execute harmonic motion and find the frequency of its oscillations.** If $d_o = 20$ cm, what is the period of the oscillations?
- Solution:
  - The forces on the bottle are mg downward, and the buoyancy force upward. Do you recall how to determine the buoyancy force?
  - The **volume of displaced water** is Ad_o, where A is the cross-sectional area of the bottle. The **mass of the displaced volume** is thus ρAd_o, and **the weight is** ρgAd_o**.** At this depth, we have $\boxed{mg = \rho gAd_o.}$
  - The **equation of motion**, then, is $\boxed{m\ddot{x} = mg - \rho gA(d_o + x),}$ where we have used $d = d_o + x$ (i.e. x is measured from the equilibrium position).

The buoyance force is equal to the weight of the displaced water. Archimedes' Principle



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Example 5.2, cont'd

- Solution, cont'd:
  - Since the weight of the bottle is the same as the displaced water at equilibrium:
    $$mg = \rho g A d_o.$$
    we can replace mg and get
    $$\boxed{\ddot{x} = -\rho g A x / m,}$$
    but we can express $\rho g A / m = g / d_o$, so we end up with the simple result
    $$\boxed{\ddot{x} = -\frac{g}{d_o} x,}$$
  - This is the same form as the simple harmonic oscillator, with the same solution. We identify
    $$\omega = \sqrt{\frac{g}{d_o}},$$
  - You may recognize this as the same expression as for a pendulum of length $l = d_o$. Note that the result does not depend on the shape or weight of the bottle, or the density of the water, or anything else. If $d_o = 20$ cm, then
    $$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{d_o}{g}} = 0.9 \text{ s}.$$

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Energy Considerations

- We have done these problems considering the equation of motion, i.e. using Newton's 2nd Law. We can also consider what happens from an energy point of view.
- Considering again the problem of a cart on a spring, with $x(t) = A\cos(\omega t - \delta)$, we have a potential energy
  $$\boxed{U = \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2\cos^2(\omega t - \delta).}$$
- Differentiating x to get the velocity, we find the kinetic energy is
  $$\tfrac{1}{2}m\dot{x}^2 = \tfrac{1}{2}m\omega^2 A^2 \sin^2(\omega t - \delta)$$
  $$\boxed{T = \tfrac{1}{2}kA^2\sin^2(\omega t - \delta),}$$
  where we have used the fact that $\omega^2 = k/m$.
- Then the total energy is $\boxed{E = T + U = \tfrac{1}{2}kA^2}$. What happens is that the kinetic energy (going as $\sin^2$) is 180 degrees out of phase with the potential energy (which goes as $\cos^2$). When U is maximum, T is minimum, and vice versa. When added together, the $\cos^2\theta + \sin^2\theta = 1$, so total energy is constant.

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Problem 5.8: Mass on a Spring

- Statement of the problem (a):
  - (a) If a mass $m = 0.2$ kg is tied to one end of a spring whose force constant $k = 80$ N/m and whose other end is held fixed, what are the angular frequency ω, the frequency f, and the period τ of its oscillations?
- Solution (a):
  - The angular frequency for a mass on a spring is given by $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{80\ \text{N/m}}{0.2\ \text{kg}}} = 20\ \text{s}^{-1}.$
  - The corresponding frequency is $f = \frac{\omega}{2\pi} = 3.2\ \text{Hz}.$
  - The period is $\tau = \frac{1}{f} = 0.31\ \text{s}.$
- Statement of the problem (b):
  - (b) If the initial position and velocity are $x_o = 0$ and $v_o = 40$ m/s, what are the constants A and δ in $x(t) = A\cos(\omega t - \delta)$?
- Solution (b): $x(0) = A\cos(-\delta) = 0 \;\Rightarrow\; \delta = \pm\frac{\pi}{2}$

  $$v(0) = \dot{x}(0) = \omega A\sin(-\delta) = \pm\omega A\sin(\tfrac{\pi}{2}) = v_o \;\Rightarrow\; A = v_o/\omega = 2\ \text{m}.$$
  - We have taken the + sign, since the velocity is positive. Thus, $\delta = -\frac{\pi}{2}$